

Answer of question no. → 02

Ans DOM :- The Document Object Model (DOM) is a programming interface for web documents that represents a web page as a structured tree of objects. It serves as a bridge that allows Javascript to connect with and manipulate the static content, structure and styling of HTML in real-time.

Core Concept of the DOM

- The Document :- The root object representing the entire web page.
- Nodes :- Every part of the document (tags, text, attributes) is a node.
- Elements :- Specific types of nodes that correspond directly to HTML tags, such as `<div>`, `<h1>` or `<p>`.

How Javascript Interacts with HTML elements

- 1) Selection (Accessing Elements) : Javascript first finds the elements it needs to change using methods like `getElementById` for unique IDs.
- 2) Manipulation (Changing Content & style) : Once selected, Javascript can modify properties such as `textContent` (updating text), `innerHTML` or `style.color`.

3) Creation & Deletion: Developers can dynamically build the page by using `document.createElement()` to add new parts or `remove()` to delete existing ones without reloading the entire page.

4) Event Handling: The DOM allows JavaScript to "listen" for user actions like clicks or keystrokes using `addEventListener`. When an event occurs, it triggers a function to update the page.